

DispatchFlow — Product Requirements Document (PRD)

1. Product Vision

DispatchFlow is an internal operations logistics platform designed for companies operating across multiple cities and provinces in Africa, especially the Democratic Republic of the Congo.

The platform enables branches outside the capital city to request items, track procurement, monitor shipment progress, and confirm delivery through a centralized operational system.

The objective is to eliminate:

- WhatsApp chaos
- Lost requests
- Missing shipment visibility
- Poor accountability
- Manual followups
- Delayed operational execution

The platform acts as a centralized operational infrastructure for procurement and inter-city logistics.

2. Core Problem

Organizations with multiple branches struggle with:

- No centralized request management
 - No procurement tracking
 - No shipment visibility
 - No operational accountability
 - Manual coordination via phone calls and WhatsApp
 - Missing proof of delivery
 - Delays without visibility
 - No analytics or operational intelligence
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3. Product Goals

Primary Goals

1. Centralize all branch requests

2. Track procurement lifecycle
 3. Track inter-city shipments
 4. Improve accountability through activity logs
 5. Reduce delays and operational confusion
 6. Provide visibility across branches
 7. Create audit trails for all operations
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4. User Types

Employee

Can:

- Create requests
 - View own requests
 - Upload attachments
 - Receive updates
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Branch Manager

Can:

- Approve/reject requests
 - View branch activity
 - Monitor deliveries
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Procurement Officer

Can:

- Purchase items
 - Update purchase status
 - Upload invoices
 - Manage suppliers
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Dispatch Officer

Can:

- Create shipments

- Update shipment progress
 - Upload transport evidence
 - Manage logistics timeline
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Regional Manager

Can:

- View regional analytics
 - Monitor operational performance
 - Track delays
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Super Admin

Can:

- Manage organizations
 - Manage permissions
 - Access all analytics
 - Configure system settings
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5. Main Workflow

Phase 1 — Request Creation

A branch creates a request.

Required Fields

- Request title
- Department
- Branch location
- Priority
- Requested items
- Quantity
- Deadline
- Budget estimate
- Notes
- Attachments

Initial Status

Pending Approval

Phase 2 — Approval

Manager reviews request.

Possible Actions:

- Approve
- Reject
- Request modifications

Statuses

- Approved
 - Rejected
 - Needs Clarification
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Phase 3 — Procurement

Procurement officer purchases items.

Required Updates

- Supplier
- Cost
- Invoice upload
- Purchase date
- Purchase notes

Status

Items Purchased

Phase 4 — Dispatch Preparation

Dispatch team prepares shipment.

Required Updates

- Transport company

- Driver information
- Waybill number
- Photos
- Expected delivery date
- Packaging notes

Status

Ready for Shipment

Phase 5 — Shipment Tracking

The shipment progresses through multiple locations.

Example: Kinshasa → Mbuji-Mayi → Kabinda

Status Examples

- In Transit
- Arrived at Transit Point
- Delayed
- Departed
- Awaiting Pickup

Timeline Requirement

Every shipment update must generate:

- Timestamp
 - User
 - Location
 - Notes
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Phase 6 — Delivery Confirmation

Branch confirms delivery.

Confirmation Options

- Fully Received
- Partially Received
- Damaged
- Missing Items

Required Attachments

- Photos
- Signature
- Notes

Final Status

Completed

6. Core Features

Authentication

- Email login
 - Phone login
 - Role-based permissions
 - Password reset
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Request Management

- Create request
 - Edit request
 - Approvals
 - Attachments
 - Priority management
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Shipment Tracking

- Real-time status updates
 - Transit points
 - Estimated delivery date
 - Delays
 - Timeline visualization
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Procurement Management

- Supplier tracking
- Cost tracking
- Invoice uploads

- Purchase history
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Notifications

WhatsApp Notifications

- Request approved
- Shipment dispatched
- Shipment delayed
- Delivery completed

Email Notifications

- Formal reports
- Delivery confirmations

SMS Notifications

- Backup communication
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Activity Logs

Every action must be recorded.

Example

- Pascal approved request
 - John updated shipment status
 - Procurement uploaded invoice
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File Uploads

Supported:

- Images
 - PDFs
 - Receipts
 - Waybills
 - Delivery signatures
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Dashboard Analytics

KPIs

- Total requests
 - Delayed shipments
 - Average delivery time
 - Most active branches
 - Procurement costs
 - Monthly operational volume
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7. Mobile Requirements

The application must be mobile-first.

Requirements

- Responsive UI
 - Fast loading
 - Camera uploads
 - Offline support
 - Progressive Web App support
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8. Non-Functional Requirements

Performance

- Page load under 3 seconds
- Real-time updates
- Optimized mobile experience

Security

- Role permissions
- Secure file uploads
- Encrypted authentication

Scalability

- Multi-company architecture
- Thousands of requests support

Reliability

- Audit logs
 - Backup systems
 - Error monitoring
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9. MVP Scope

MUST HAVE

- Authentication
- Roles and permissions
- Request creation
- Approval system
- Shipment tracking
- Timeline updates
- Notifications
- Attachments
- Dashboard
- Activity logs

DO NOT BUILD YET

- AI features
 - Advanced maps
 - ERP integrations
 - Accounting systems
 - Fleet optimization
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10. Recommended Tech Stack

Frontend

- Next.js
- Tailwind CSS
- shadcn/ui

Backend

- Supabase

Database

- PostgreSQL

File Storage

- Supabase Storage

Notifications

- WhatsApp Cloud API
- Twilio

Hosting

- Vercel
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11. Future Features

Phase 2

- Inventory management
 - Warehouse management
 - Asset tracking
 - Fleet management
 - AI delay prediction
 - Procurement forecasting
 - Multi-language support
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12. Market Positioning

DispatchFlow is not a delivery app.

It is:

- Operational infrastructure
 - Procurement workflow system
 - Internal logistics platform
 - Multi-branch operations operating system
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13. Ideal Customers

- NGOs
- Mining companies
- Telecom companies
- Schools
- Hospitals
- Construction firms
- Churches
- Distribution companies

Especially organizations operating across multiple provinces.

14. Success Metrics

Operational KPIs

- Reduced delivery delays
- Faster procurement cycles
- Increased visibility
- Reduced lost shipments
- Faster approvals

Business KPIs

- Monthly active organizations
 - Retention rate
 - Average revenue per account
 - Operational adoption rate
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15. Launch Strategy

Phase 1

Pilot with one company.

Phase 2

Deploy to multiple branches.

Phase 3

Add analytics and intelligence.

Phase 4

Expand into broader ERP infrastructure.